

SRAVYA BUDDHA

Data Scientist | Health Sciences, Non-Profits & Clinical Engineering

Passionate researcher and data specialist with 5+ years of experience scaling data operations from administrative workflows to high-dimensional biomedical pipelines. Expert in leveraging Python, SQL, and Neo4j knowledge graphs to map complex disease phenotypes and translate intricate behavioral datasets into interactive visual dashboards. A resourceful, rapid learner who excels at engineering impactful, real-world data solutions in resource-constrained environments.

CORE COMPETENCIES

Clinical & Behavioral Analytics: Disease Phenotype Mapping • Behavioral Health Informatics • Patient Data

Data Engineering & Systems: Python • SQL • Knowledge Graphs (Neo4j) • Pipeline Automation (ETL)

Operations & Reporting: Executive Dashboards • Resource Optimization • SOP Development • Tenant Reporting

PROFESSIONAL EXPERIENCE

Research Data Scientist (Knowledge Graphs) - University of Delaware, Newark, DE, Mar 2026 – Present

Serve in an active biomedical research group, engineering the data structures required to integrate, visualize, and analyze complex pharmacological and biological relationships.

Select Accomplishments:

- Discovered ~10K novel drug–protein mechanisms by leading cross-database entity resolution across Open Targets, ChEMBL, and UniProt expanding the biological coverage of the ProKN knowledge graph.
- Architected a Neo4j biomedical knowledge graph mapping behavioral health phenotypes (e.g., Major Depressive Disorder, ADHD) to genes, proteins, and compounds enabling translational research insights across clinical and molecular layers.
- Engineered data pipelines integrating 20K+ drug and protein entities using deterministic UUID v5 hashing and chemical identifiers (InChIKey, ChEMBL, PubChem), ensuring zero-duplication and high-confidence entity resolution.
- Optimized graph performance and data integrity by designing ETL transformations to deduplicate high-volume biological relationships and implementing schema constraints to prevent orphan nodes and invalid edges.

Digital Strategy & Analytics Intern - University of Delaware, Newark, DE, Jan 2025 – Present

Manage data operations, reporting, and digital strategy for a multi-tenant research facility. Act as the primary liaison between facility tenants and executive leadership, optimizing workflows using lean, sustainable resources.

Select Accomplishments:

- Enabled a \$50M grant funding pipeline by designing a structured tenant feedback and evaluation framework, translating qualitative research inputs into actionable metrics for funding success.
- Designed an AI-driven operational system to automate SOP generation across 20+ laboratories, improving compliance consistency and reducing manual documentation overhead.
- Analyzed and reported tenant engagement and operational data to executive leadership, directly informing facility strategy and ecosystem growth decisions.

- Designed scalable digital infrastructure supporting a 40-company scientific ecosystem, aligning operational workflows with funding, compliance, and research execution needs.
- Led data literacy and analytics training (Power BI, AI workflows) for 20+ research teams, achieving 60% sustained adoption and improving data-driven decision-making across tenants.

Data Engineer/Analyst - Six Sigma Engineering Enterprises, India, Jan 2022 – Oct 2023

Led data modeling and analytics initiatives for a product-based engineering firm. Directed data-driven marketing strategies, managed customer retention models, and optimized ad campaigns based on continuous traffic and sales analytics.

Select Accomplishments:

- Generated ₹500,000 in additional monthly revenue by developing predictive models to forecast customer purchasing behavior and trigger targeted retention campaigns.
- Retained ~15–20 customers/month by implementing data-driven retention strategies improving customer lifecycle value.
- Improved inventory forecasting accuracy by 15%, reducing overstock costs by ₹200,000 through ETL pipelines.

Founder (Data & Strategy Lead)- My Kiwi Butter, New Zealand, Jun 2020 – Dec 2021

Spearheaded the launch and scalability of a data-centric D2C enterprise, orchestrating comprehensive commercial strategies and localized informatics while driving growth through iterative behavioral feedback loops.

Select Accomplishments:

- Increased conversion rates by 36% by building SQL & PowerBI customer segmentation and pricing optimization models.
- Established a recurring customer base of 100+ users by designing automated, data-driven marketing workflows.

EDUCATION

Master of Science in Data Science (STEM OPT Eligible) – University of Delaware, Expected May 2026 - GPA: 3.8